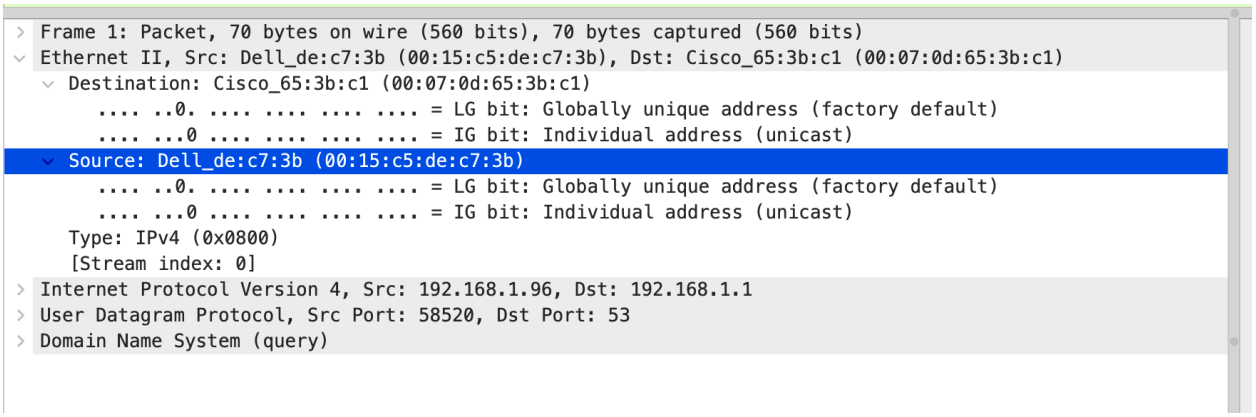
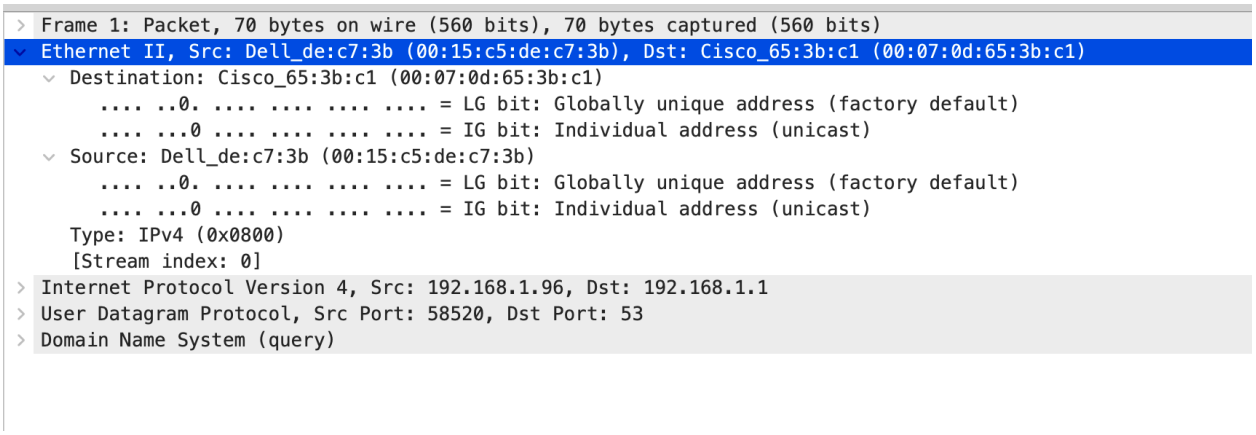
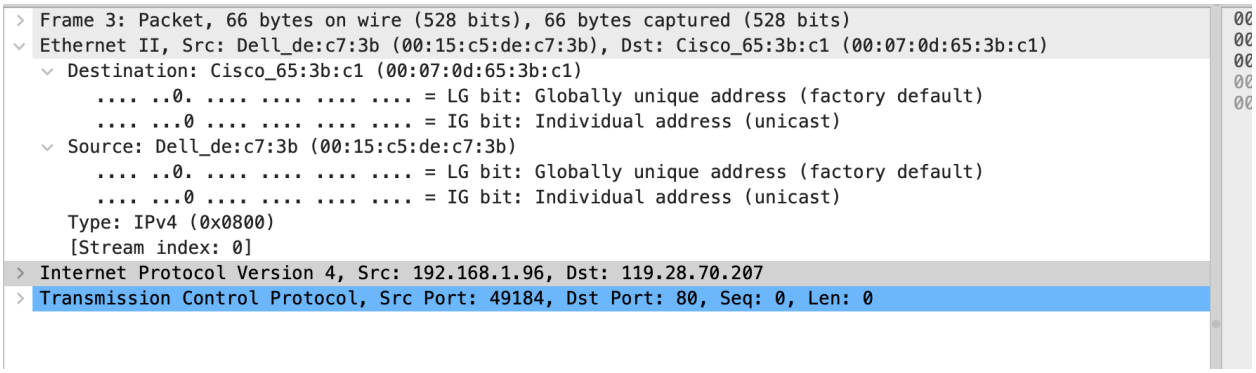
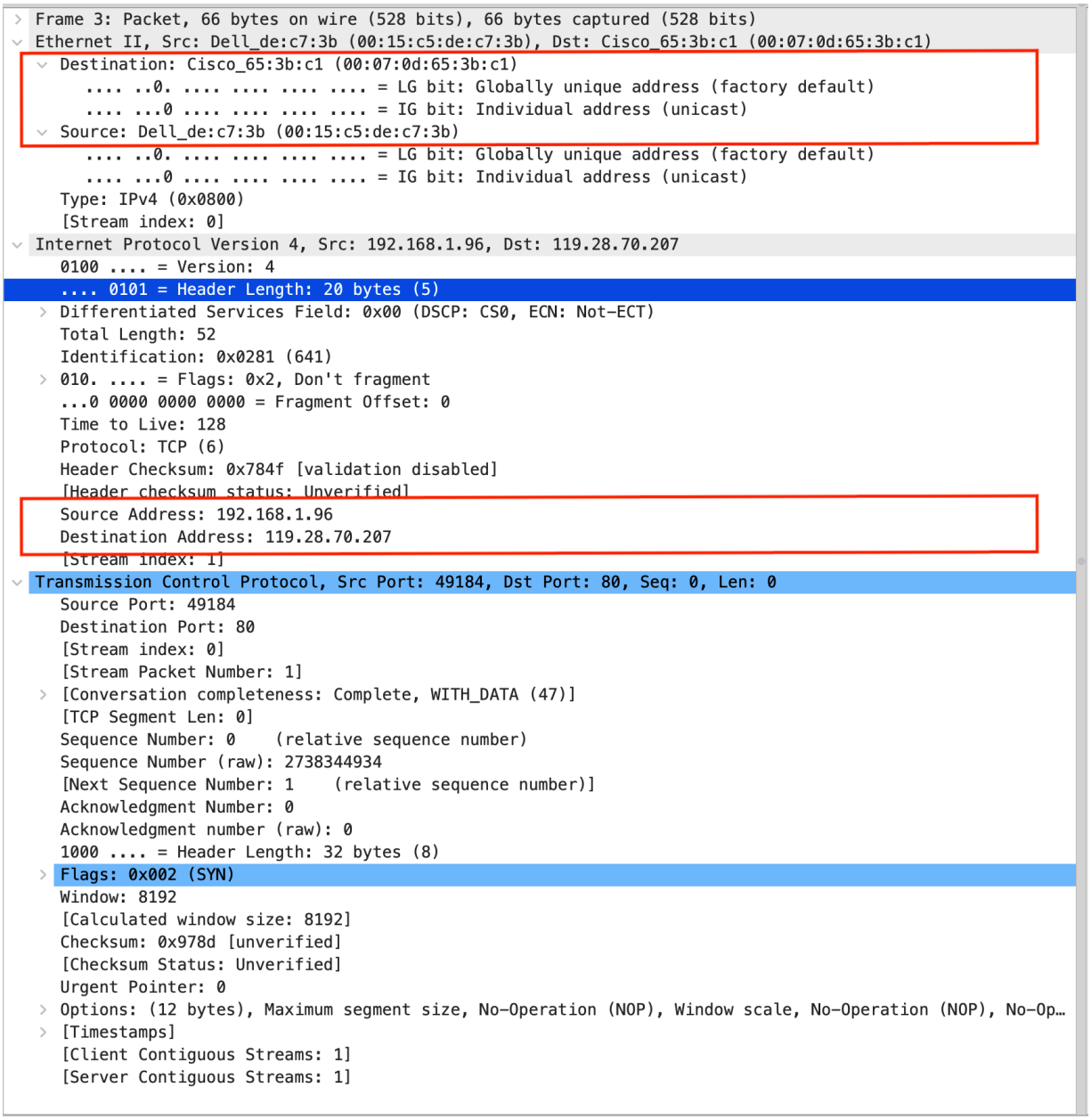




Identified the source machine Dell_de:c7:3b in Frame 1, with the hypothesis that the dell node was a workstation, and that the cisco node was a router. Set about proving this using other information in subsequent packets.



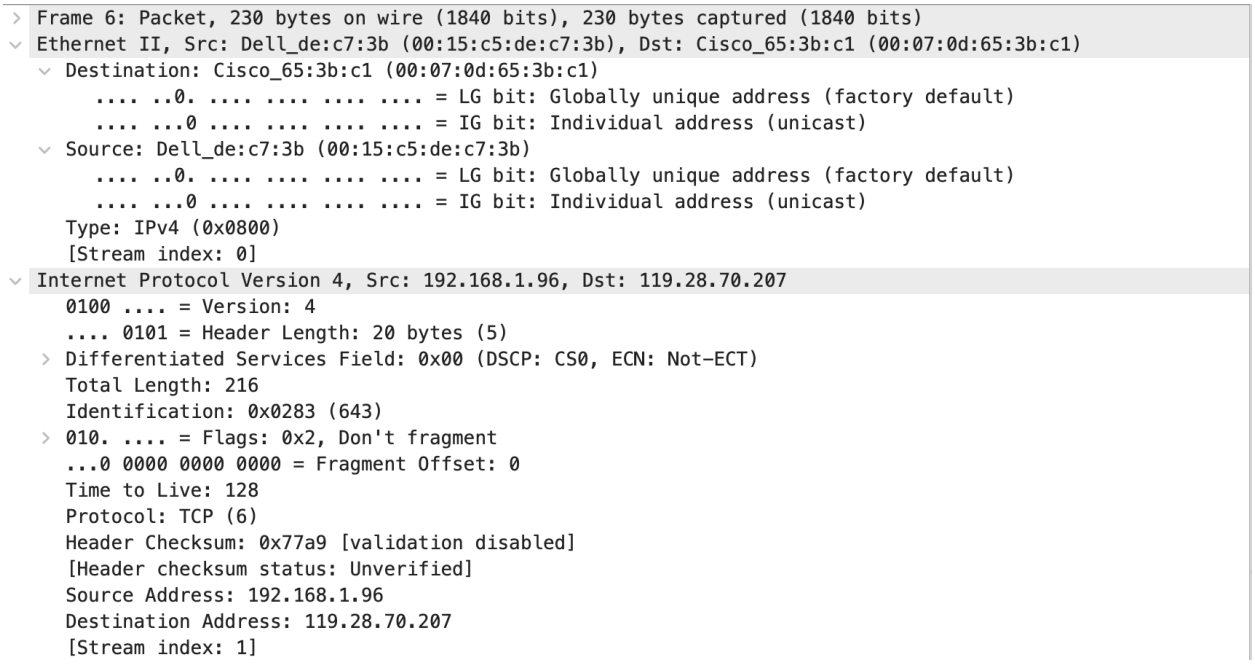
Packet 3 shows the same Dell node, IP ending .96, communicating with another ipv4 address over the web – supporting my hypothesis that the dell node is a workstation within the network, communicating with another device over the internet.



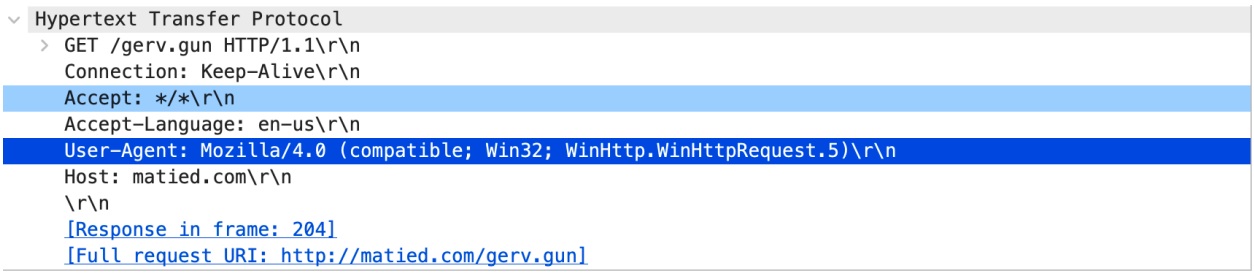
The above screenshot shows the key pieces of information used to support the hypothesis that the dell machine is a workstation, communicating with another device other the internet, with the cisco machine being a router.

Under the ethernet tab the cisco device is shown as the destination on the internal network, with the dell as the source. Under the IP headers, the source IP address is the address of the dell machine and the destination address is an ipv4 address on the internet, showing that the dell is communicating with another machine over the internet.

This shows that the cisco is the router in this situation given that a node inside of a network can only directly address (via mac address) other nodes within the same network. To address a machine over the internet, the request needs to go through the default gateway, i.e. the router.



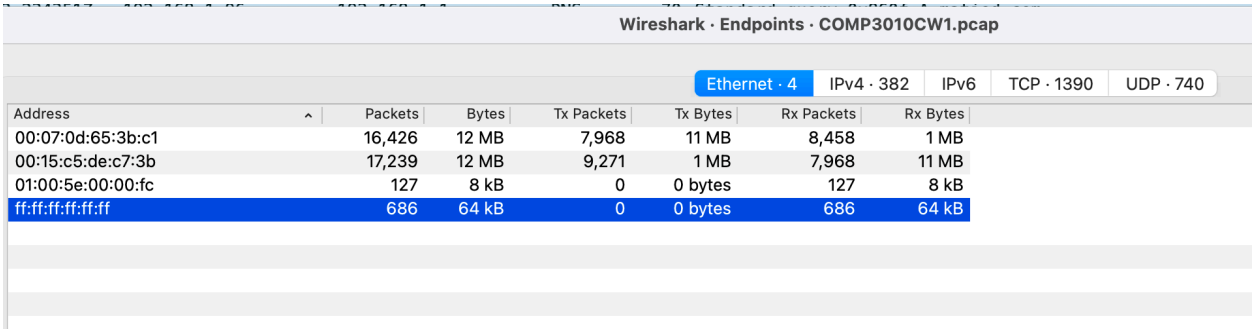
Packet 6, in the above screenshot, shows the same ethernet source and destination – again suggesting a hop between the dell as the windows machine, and cisco as the router. Given that this is a HTTP request, we have some additional information that can be used.



The user agent in the HTTP request includes Win32 and WinHttp.WinHttpRequest.5. WinHttpRequest is a HTTP client used within windows applications[i] to send requests over the HTTP protocol to other HTTP servers.

In addition, the TTL is 128 – which is the windows default[ii]. This is weak evidence on its own, but does strengthen the above.

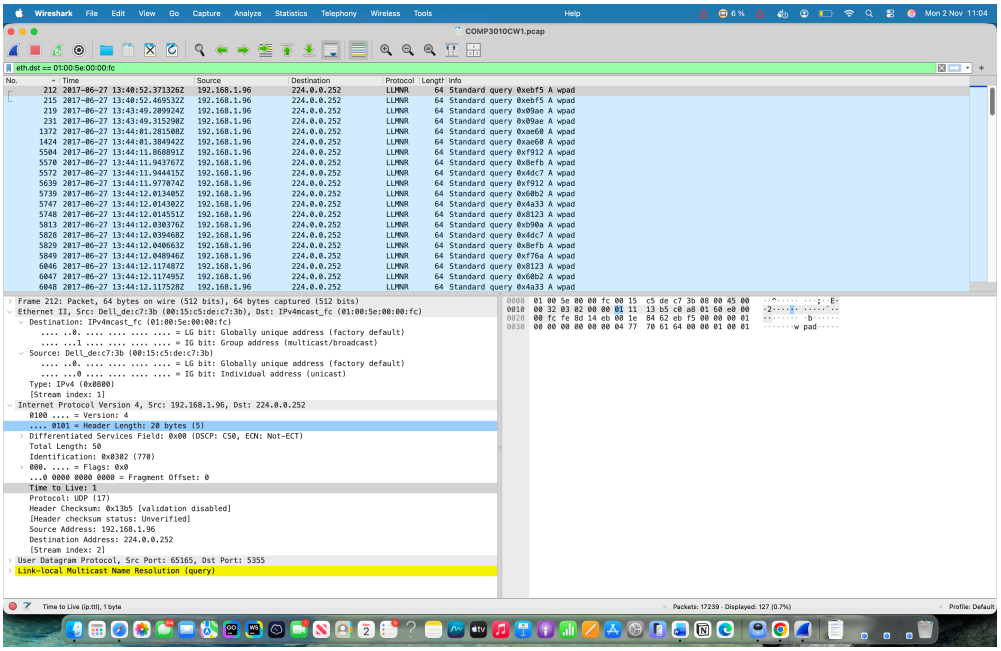
Before reaching the conclusion that the priorly referenced Dell machine is the vulnerable windows machine, I needed to rule out any other possible machines.



Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
00:07:0d:65:3b:c1	16,426	12 MB	7,968	11 MB	8,458	1 MB
00:15:c5:de:c7:3b	17,239	12 MB	9,271	1 MB	7,968	11 MB
01:00:5e:00:00:fc	127	8 kB	0	0 bytes	127	8 kB
ff:ff:ff:ff:ff:ff	686	64 kB	0	0 bytes	686	64 kB

The above screenshot shows all endpoints detected in the PCAP file on ethernet.

1. 00:07:0d:65:3b:c1 – the cisco device, which is likely the router as determined above.
2. 00:15:c5:de:c7:3b – The dell machine suspected of being our windows machine
3. 01:00:5e:00:00:fc –

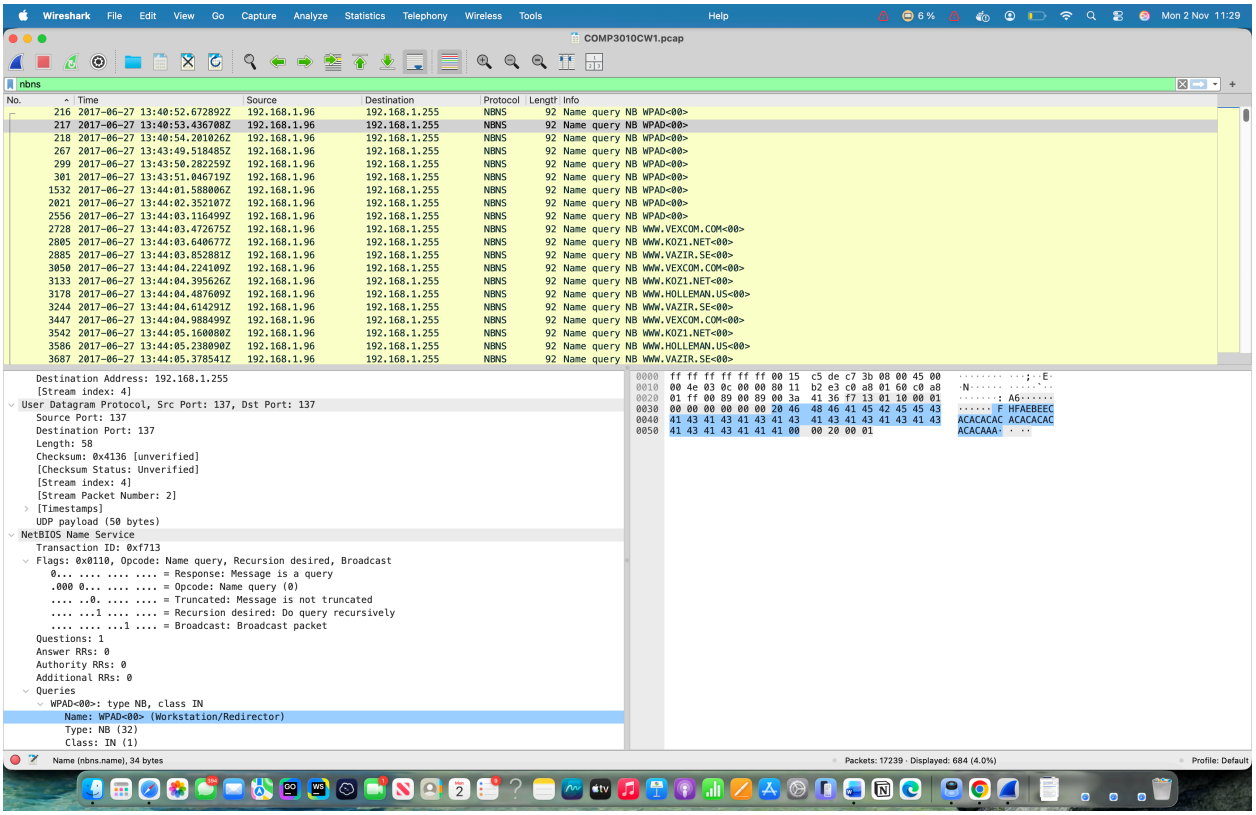


I inititally suspected that this was DNS lookups, or something similar. Upon further investigation I identified this as LLMNR, which enables name resolution in scenarios in which conventional DNS name resolution is not possible, B. Adoba, D.Thaler, et. al[iii]

4. ff:ff:ff:ff:ff:ff – the default broadcast mac address, which is a destination – not a machine.

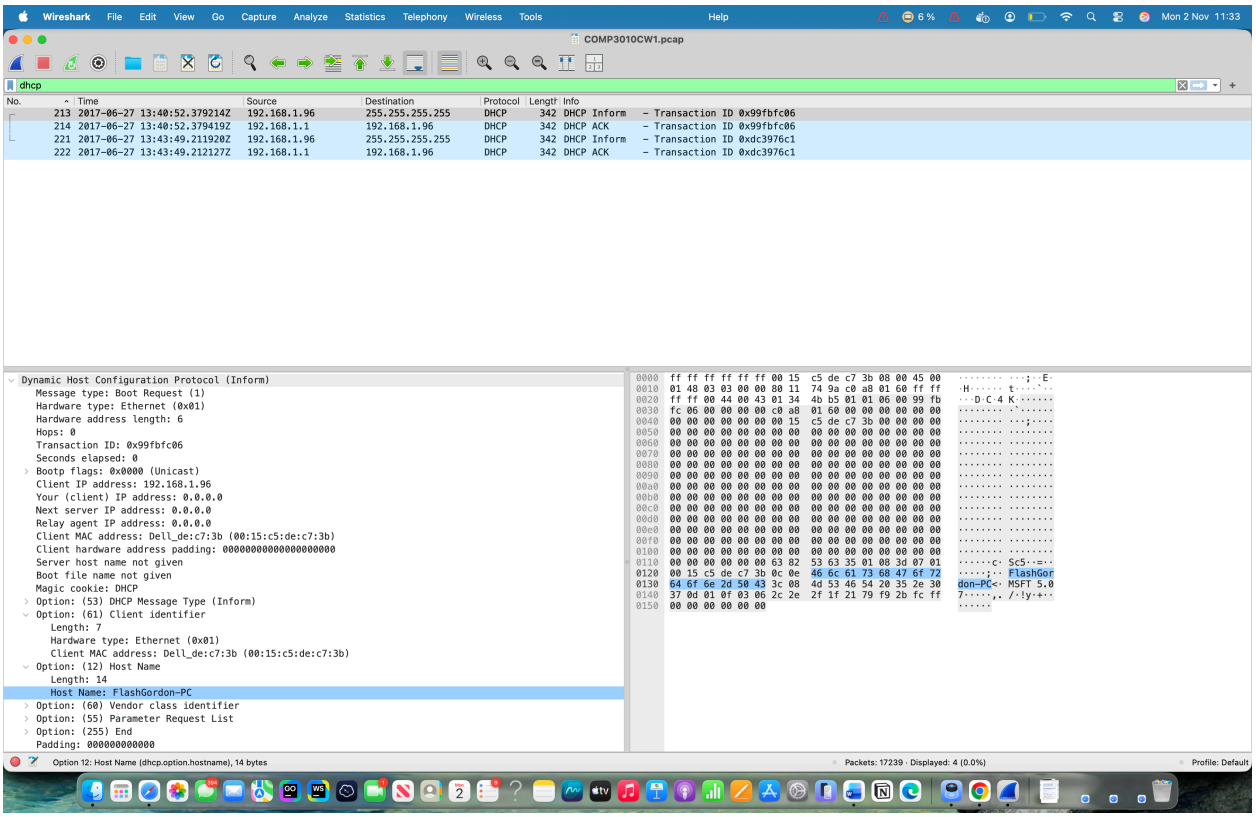
The above shows that two of the four ethernet rows are destination-only, leaving only the Dell and Cisco machines. This supports my findings given that no other hosts appear in the captured traffic, so there is nothing in the data to contradict my identification of the Dell as the infected windows machine.

Hostname



Checked for NBNS packets but only found outbound name queries

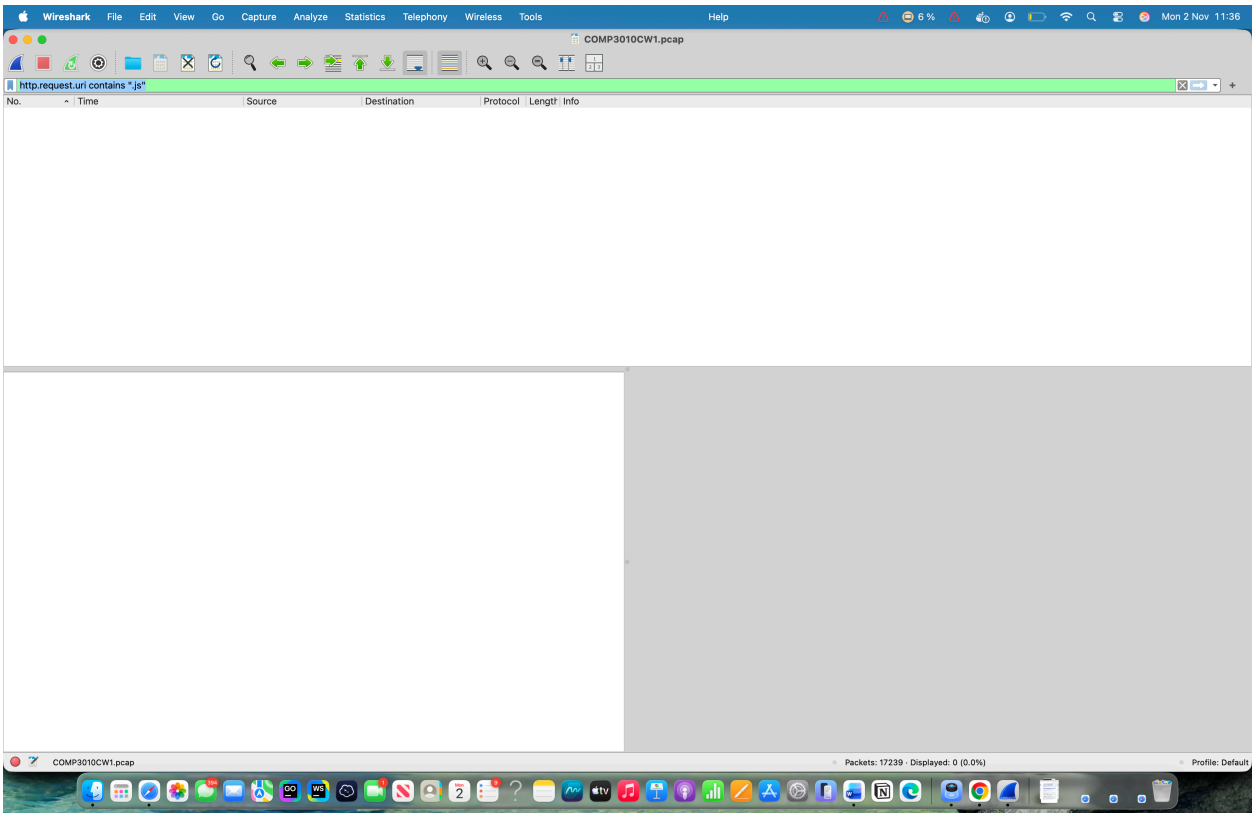
Checked DHCP since machines identify themselves with Mac Adress and hostname to be assigned an IP:



Found the DHCP inform call from the dell machine, which exposes the host name

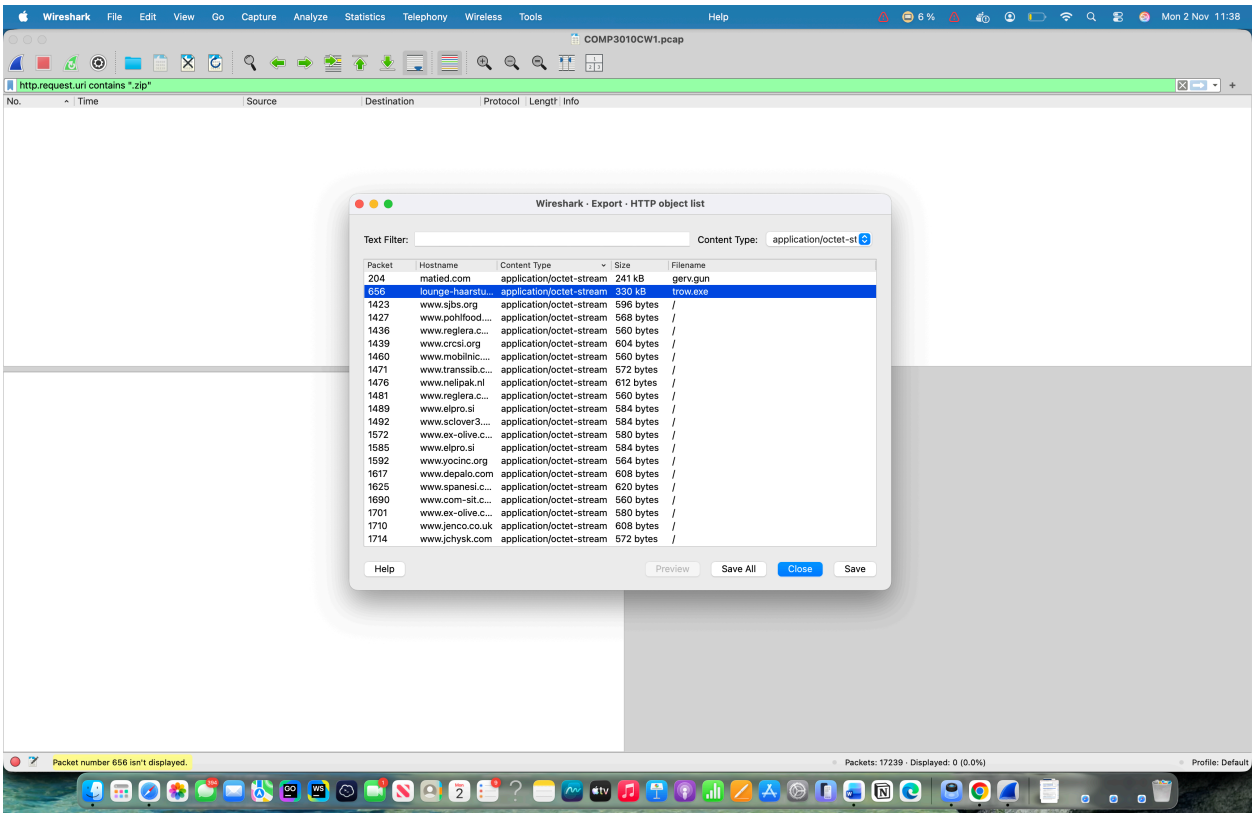
Date of infection.

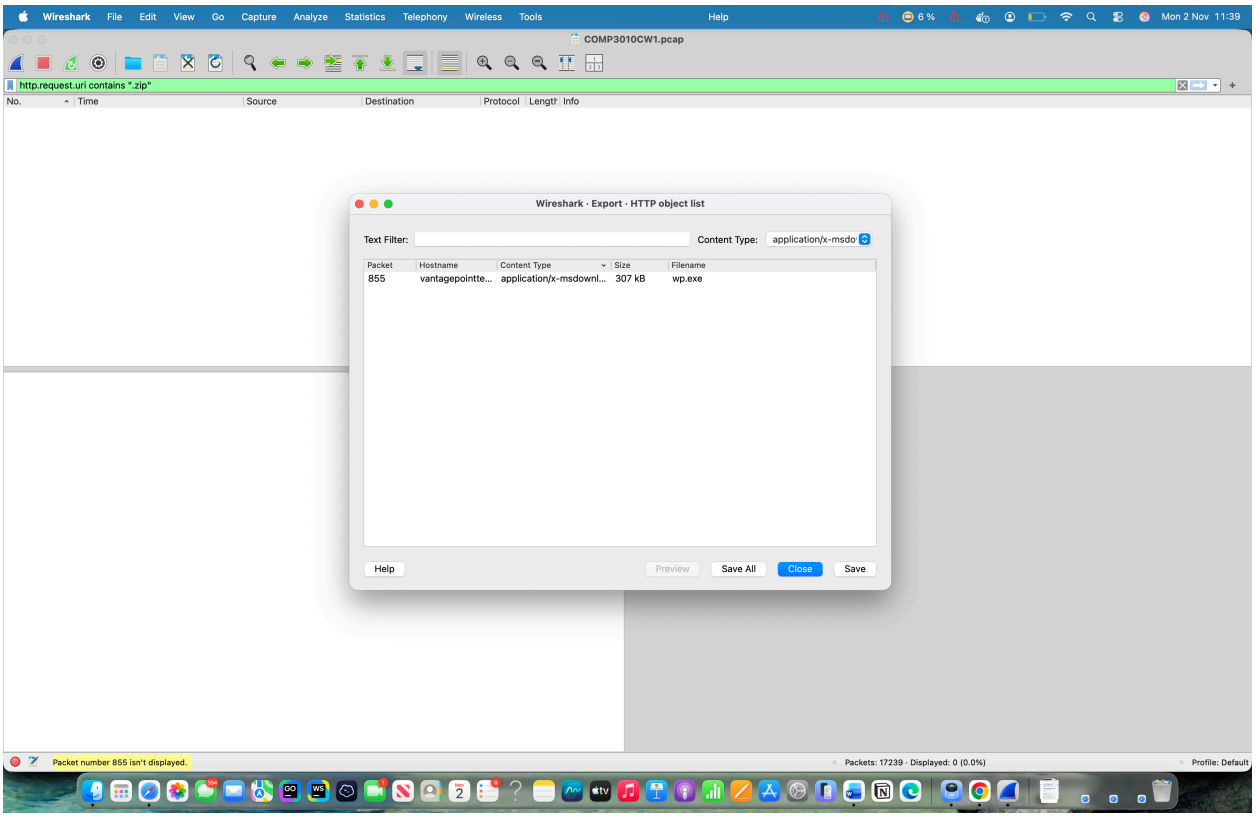
Given that there is a mention of the infection “probably being caused by a js file”, I first started with a search for a .js file, which turned up no results (as below):

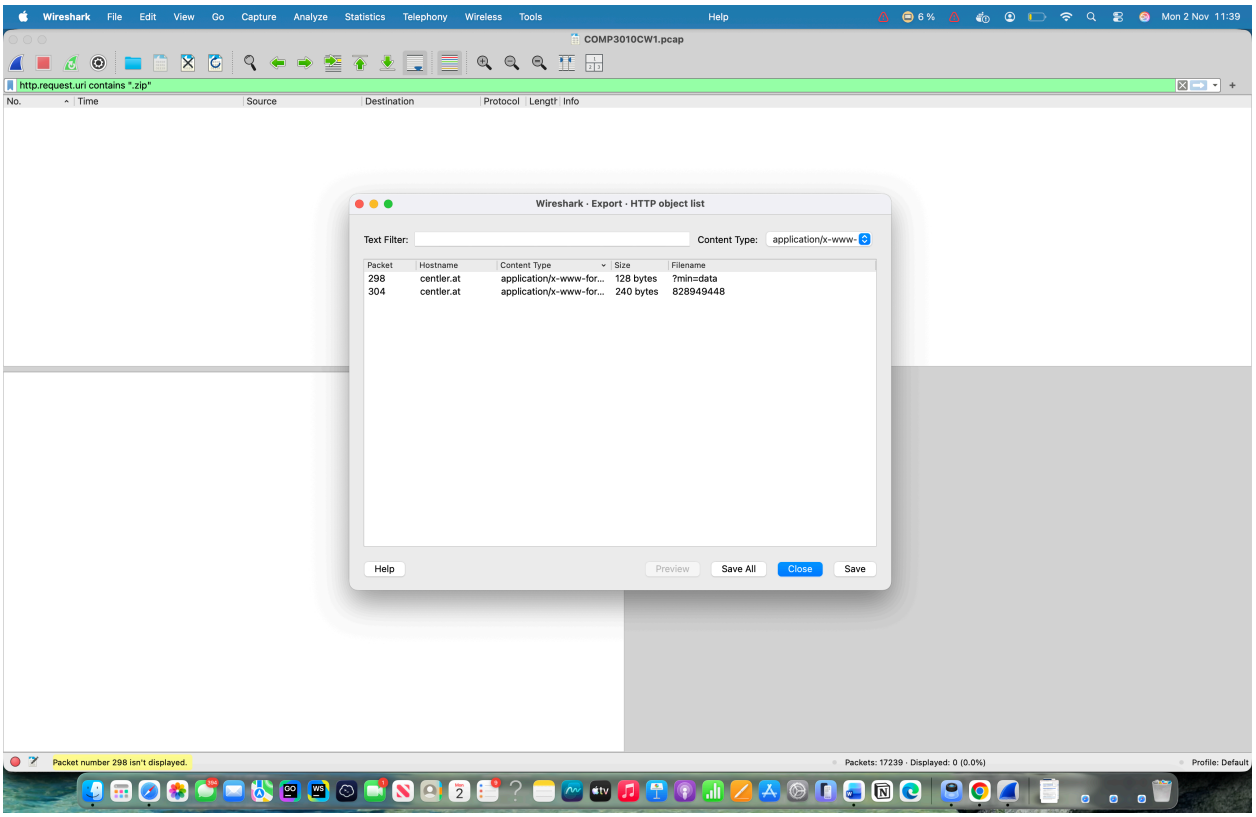


I also did the same for .zip and got nothing back.

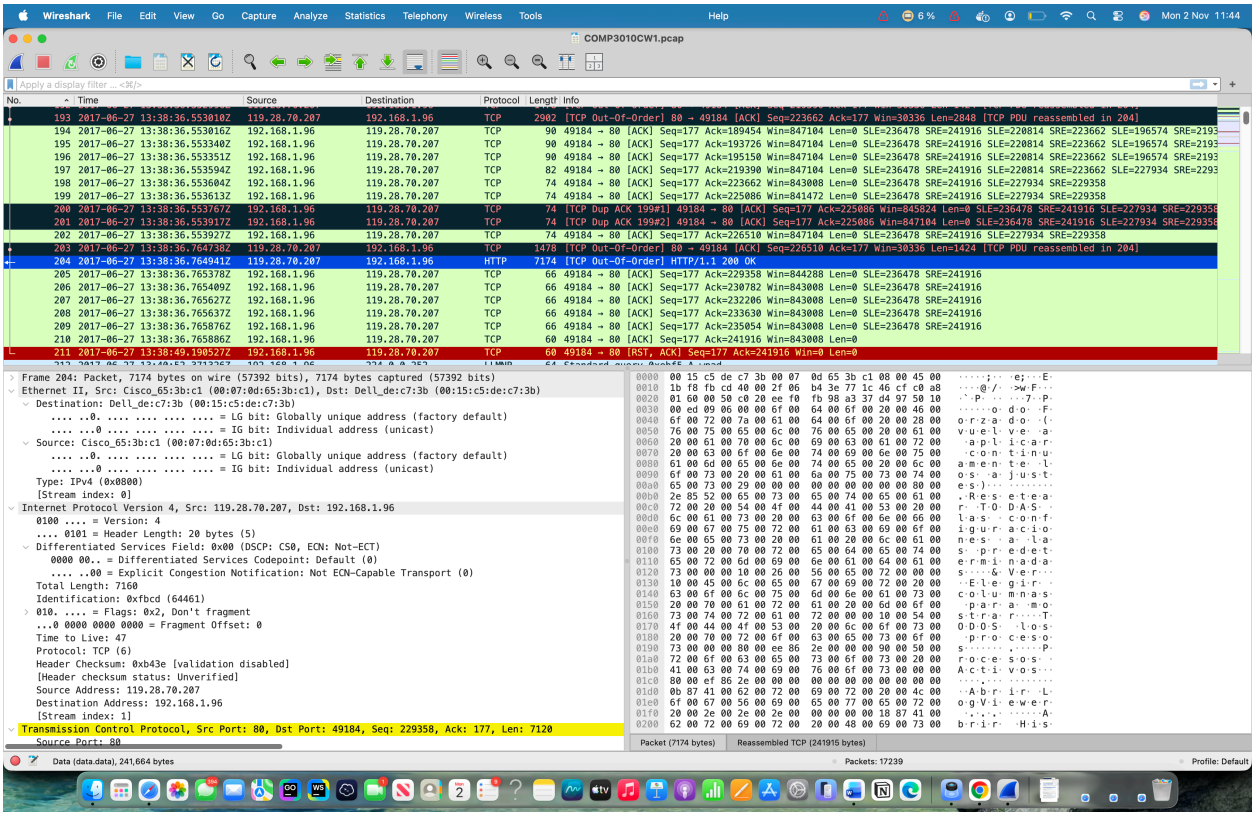
Upon checking File > Export Objects > HTTP


Wireshark File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Mon 2 Nov 11:38
COMP3010CW1.pcap
http.request.uri contains ".zip"
No. Time Source Destination Protocol Length Info
Wireshark · Export · HTTP object list
Text Filter:
Content Type: application/octet-st
Packet Hostname Content Type Size Filename
204 matied.com application/octet-stream 241 kB gerv.gun
656 lounge-haarstu... application/octet-stream 330 kB trow.exe
1423 www.sjbs.org application/octet-stream 596 bytes /
1427 www.pohlfood.... application/octet-stream 568 bytes /
1436 www.reglera.c... application/octet-stream 560 bytes /
1439 www.crcsi.org application/octet-stream 604 bytes /
1460 www.mobilnic.... application/octet-stream 560 bytes /
1471 www.transsib.c... application/octet-stream 572 bytes /
1476 www.nelipak.nl application/octet-stream 612 bytes /
1481 www.reglera.c... application/octet-stream 560 bytes /
1489 www.elpro.si application/octet-stream 584 bytes /
1492 www.sclover3.... application/octet-stream 584 bytes /
1572 www.ex-olive.c... application/octet-stream 580 bytes /
1585 www.elpro.si application/octet-stream 584 bytes /
1592 www.yocinc.org application/octet-stream 564 bytes /
1617 www.depalo.com application/octet-stream 608 bytes /
1625 www.spanesi.c... application/octet-stream 620 bytes /
1690 www.com-sit.c... application/octet-stream 560 bytes /
1701 www.ex-olive.c... application/octet-stream 580 bytes /
1710 www.jenco.co.uk application/octet-stream 608 bytes /
1714 www.jchysk.com application/octet-stream 572 bytes /
Help
Preview
Save All
Close
Save
Packet number 656 isn't displayed.
Packets: 17239 · Displayed: 0 (0.0%)
Profile: Default


Wireshark File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Mon 2 Nov 11:39
COMP3010CW1.pcap
http.request.uri contains ".zip"
No. Time Source Destination Protocol Length Info
Wireshark · Export · HTTP object list
Text Filter:
Content Type: application/x-msdo
Packet Hostname Content Type Size Filename
855 vantagepointte... application/x-msdownl... 307 kB wp.exe
Help
Preview
Save All
Close
Save
Packet number 855 isn't displayed.
Packets: 17239 · Displayed: 0 (0.0%)
Profile: Default



Of the three downloads, gerv.gun was the earliest – with packet number 204. I need to work backwards from this packet to find the initiator of the download:



1	2017-06-27 13:38:32.234351Z	192.168.1.96	192.168.1.1	DNS	70	Standard query 0x860f A matied.com
2	2017-06-27 13:38:32.435448Z	192.168.1.1	192.168.1.96	DNS	86	Standard query response 0x860f A matied.com A 119.28.70.207
3	2017-06-27 13:38:32.439170Z	192.168.1.96	119.28.70.207	TCP	66	49184 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
4	2017-06-27 13:38:32.651272Z	119.28.70.207	192.168.1.96	TCP	66	80 → 49184 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1424 SACK_PERM WS=128
5	2017-06-27 13:38:32.651777Z	192.168.1.96	119.28.70.207	TCP	60	49184 → 80 [ACK] Seq=1 Ack=1 Win=66816 Len=0
6	2017-06-27 13:38:32.652026Z	192.168.1.96	119.28.70.207	HTTP	230	GET /gerv.gun HTTP/1.1

Packets 1-5 show a DNS lookup/resolution, followed by a TCP handshake with the resolved IP address. Then packet 6 shows the download/GET request to /gerv.gun.

As I identified earlier, the user agent of the request in packet 6 suggests the request was made by a program running on windows, using WinHttp.WinHttpRequest.5. In addition, there is no evidence in the network capture to suggest that the user navigated to the download page via a browser – given there are no HTTP requests, with HTML responses. This solidifies the position that a program likely initiated the download of gerv.gun

Given that the captured network traffic doesn't show anything before the DNS lookup for matied.com, we cannot guarantee what initiated the download. With the evidence we have within the network capture, it seems that the moment of infection is the initiation of download – which starts at the point of the DNS lookup for matied.com

Date: 2017-06-27, Time: 13:38

Q6:

Collapsed the list to remove duplicates in terminal, then read through the headlines to identify the most suspicious logs – which I have pasted below (used a snort pretty formatting plugin for readability)

```
[**] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.127561 192.168.1.96:49326 -> 85.13.131.20:80
TCP TTL:46 TOS:0x0 ID:58745 IpLen:20 DgmLen:1351 DF
***A**** Seq: 0x54E99FD0  Ack: 0x9A752BEF  Win: 0x7F00  TcpLen: 20
```

```
[**] [1:15306:22] FILE-EXECUTABLE Portable Executable binary file magic detected [**]
[Classification: Potential Corporate Privacy Violation] [Priority: 1]
06/27-13:38:34.111294 119.28.70.207:80 -> 192.168.1.96:49184
TCP TTL:47 TOS:0x0 ID:64228 IpLen:20 DgmLen:1464 DF
***A**** Seq: 0xEEED7BAB  Ack: 0xA337D497  Win: 0xED  TcpLen: 20
```

```
[**] [1:15168:13] INDICATOR-COMPROMISE Suspicious .ru dns query [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:11.994287 192.168.1.96:51688 -> 192.168.1.1:53
UDP TTL:128 TOS:0x0 ID:3326 IpLen:20 DgmLen:54
Len: 26
```

```
[**] [1:11192:20] FILE-EXECUTABLE download of executable content [**]
[Classification: Potential Corporate Privacy Violation] [Priority: 1]
06/27-13:43:52.407756 145.131.10.21:80 -> 192.168.1.96:49190
TCP TTL:128 TOS:0x0 ID:835 IpLen:20 DgmLen:17427 DF
***A**** Seq: 0xA90C3547  Ack: 0xCB5AB7FC  Win: 0xFD20  TcpLen: 20
[Xref => http://www.microsoft.com/smallbusiness/resources/technology/security/practice_safe_computing_and_thwart_online_thugs.mspx]
```

```
[**] [1:28629:6] INDICATOR-OBFUSCATION obfuscated script encoding detected [**]
[Classification: Misc activity] [Priority: 3]
06/27-13:44:01.624079 148.251.33.194:80 -> 192.168.1.96:49194
TCP TTL:43 TOS:0x0 ID:37796 IpLen:20 DgmLen:11720 DF
***A**** Seq: 0xA2F1C810  Ack: 0x95857EE1  Win: 0x7F  TcpLen: 20
```

```
[**] [1:28406:1] MALWARE-CNC Win.Trojan.Kazy variant outbound connection [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:43:54.128138 192.168.1.96:49191 -> 143.95.151.192:80
TCP TTL:49 TOS:0x8 ID:1431 IpLen:20 DgmLen:188 DF
***A**** Seq: 0xA88DB23D  Ack: 0x2FD3568D  Win: 0x3E00  TcpLen: 20
[Xref => http://www.virustotal.com/en/file/a064a1d3d8b9d8ab649686b7fb01e0631e569412388084f5c391722c98660763/analysis/]
```

```
--------------------------------------------------------------------------------
Count:1 Event#3.49114 2017-06-28 00:20:06
ET TROJAN Backdoor.Win32.Pushdo.s Checkin
192.168.1.96 -> 54.164.249.255
IPVer=4 hlen=5 tos=0 dlen=871 ID=0 flags=0 offset=0 ttl=0 chksum=50405
Protocol: 6 sport=49274 -> dport=80

Seq=0 Ack=0 Off=5 Res=0 Flags******** Win=0 urp=13791 chksum=0
--------------------------------------------------------------------------------
```

```
--------------------------------------------------------------------------------
Count:1 Event#3.49001 2017-06-28 00:14:33
ET POLICY Binary Download Smaller than 1 MB Likely Hostile
119.28.70.207 -> 192.168.1.96
IPVer=4 hlen=5 tos=0 dlen=1464 ID=0 flags=0 offset=0 ttl=0 chksum=13645
Protocol: 6 sport=80 -> dport=49184

Seq=0 Ack=0 Off=5 Res=0 Flags******** Win=0 urp=19868 chksum=0
--------------------------------------------------------------------------------
```

```
--------------------------------------------------------------------------------
Count:1 Event#3.49054 2017-06-28 00:20:01
ETPRO TROJAN Sinkhole.tech Sinkhole Reply
144.217.254.90 -> 192.168.1.96
IPVer=4 hlen=5 tos=0 dlen=341 ID=0 flags=0 offset=0 ttl=0 chksum=26727
Protocol: 6 sport=80 -> dport=49227

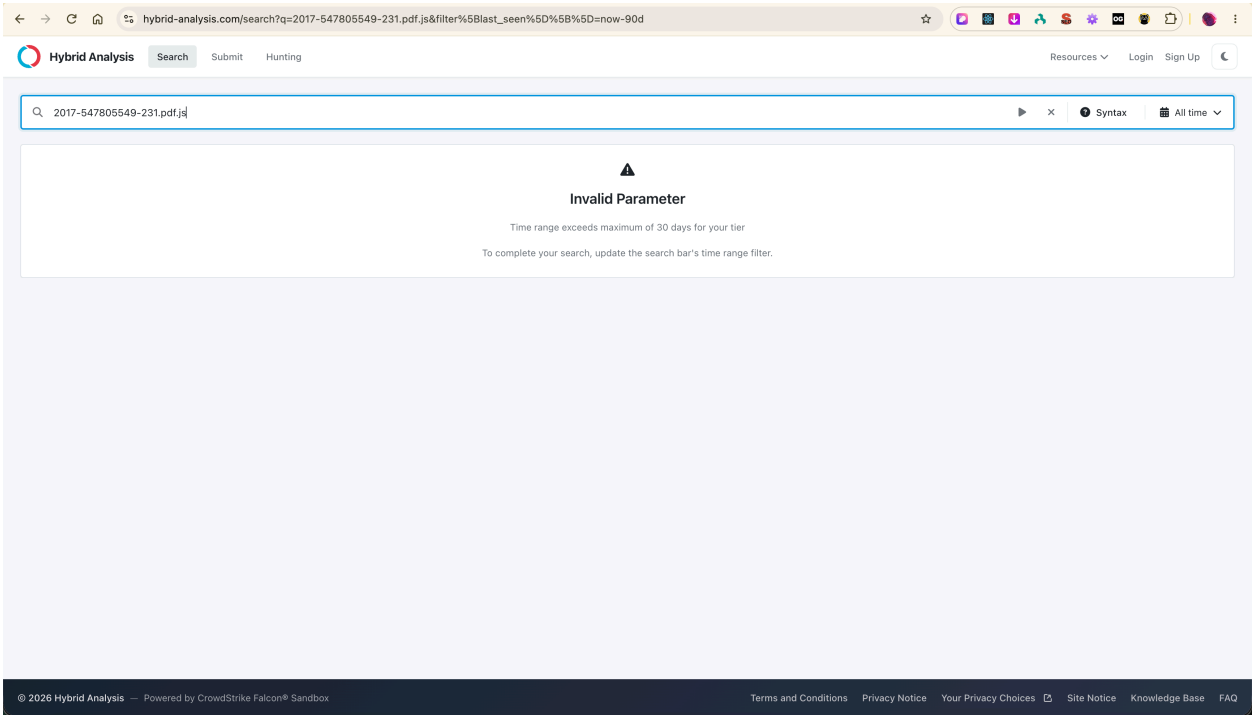
Seq=0 Ack=0 Off=5 Res=0 Flags******** Win=0 urp=54246 chksum=0
--------------------------------------------------------------------------------
```

```
--------------------------------------------------------------------------------
Count:1 Event#3.49056 2017-06-28 00:20:01
ET TROJAN Backdoor.Win32.Pushdo.s Checkin
192.168.1.96 -> 82.201.61.230
IPVer=4 hlen=5 tos=0 dlen=885 ID=0 flags=0 offset=0 ttl=0 chksum=25804
Protocol: 6 sport=49195 -> dport=80

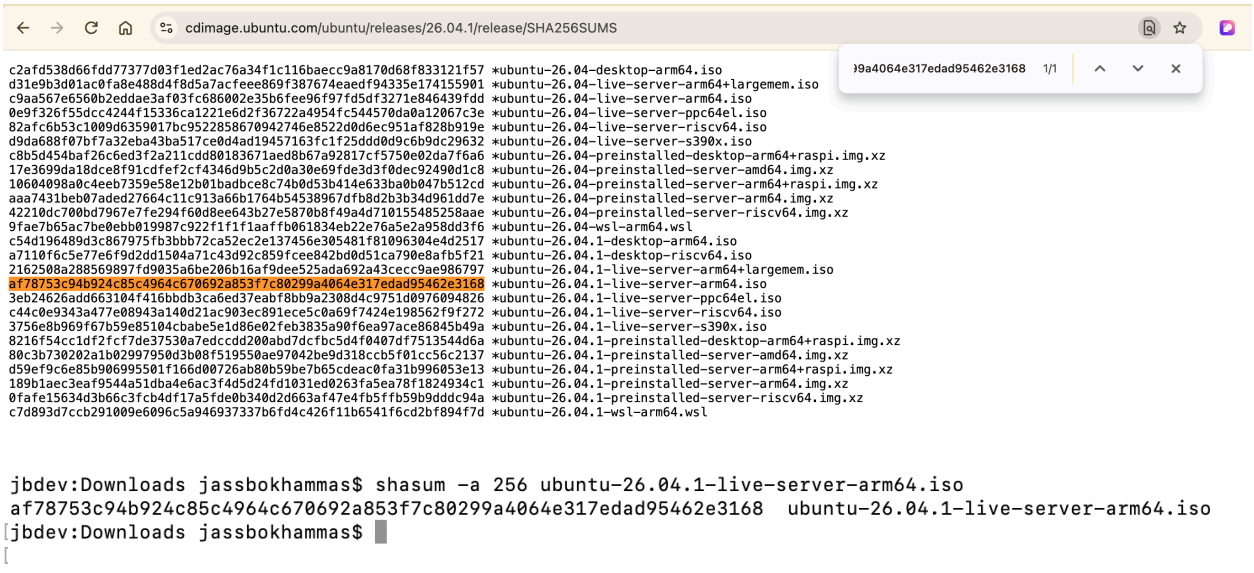
Seq=0 Ack=0 Off=5 Res=0 Flags******** Win=0 urp=6948 chksum=0
--------------------------------------------------------------------------------
```

Both sources name Pushdo, with snort naming “Win.Trojan.Pushdo” and Suricata naming “Backdoor.Win32.Pushdo.s”, whilst both tools classify the pushdo trojan differently they still agree on the variant which made the identification of the Pushdo variant more credible

given that two sources independently identified the same variant on the same network capture, with no evidence in the network capture analysis to suggest otherwise.



https://fireflyframer.blog.jp/19064394.html#:~:text=%5BScript%20Files%5D,64ae13592c5d8b289be81ebb50e054cef49b0bdf50ea92dd44ed611dd274150f/analysis/1498523440/



[i] https://learn.microsoft.com/en-us/windows/win32/winhttp/winhttp-start-page#:~:text=Microsoft%20Windows%20HTTP%20Services%20(WinHTTP)%20provides%20developers%20with%20an%20HTTP%20client%20application%20programming%20interface%20(API)%20to%20send%20requests%20through%20the%20HTTP%20protocol%20to%20other%20HTTP%20servers.

[ii] https://learn.microsoft.com/en-us/dotnet/api/microsoft.powershell.commands.testconnectioncommand.maxhops?view=powershellsdk-7.6.0#:~:text=The%20default%20(from%20Windows)%20is%20128%20hops.

[iii] https://www.rfc-editor.org/info/rfc4795/